

• mikrosk. last./obnašanje delcev $\rightarrow$ makrosk. opazljivke ($p, U, \vec{M}, \dots$)

• statistični ansambl:

- 1) nabor mikrosk. stanj sistema
- 2) frekvence teh stanj

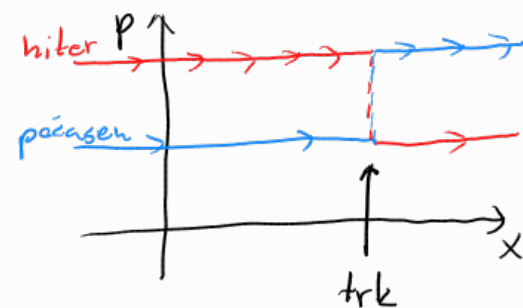
Fazni prostor

zgled: enoatomni idealni plin

- N delcev
- $3N$ koordinat: $x_1, y_1, z_1, \dots$
- $3N$ komp. $\vec{p}$: $p_{x1}, p_{y1}, p_{z1}, \dots$

$\Rightarrow 6N$ prost. stopenj

zgled: dva delca plina

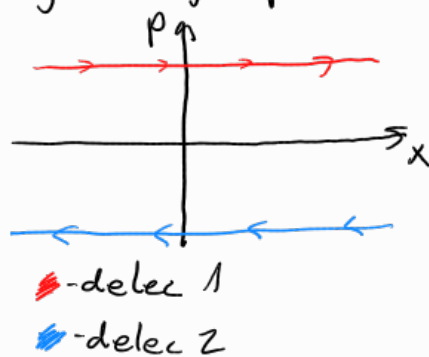


■ - delec 1

■ - delec 2

- identiteta delcev ni bistvena

zgled: mirujoči plin ($\vec{v} = 0$)



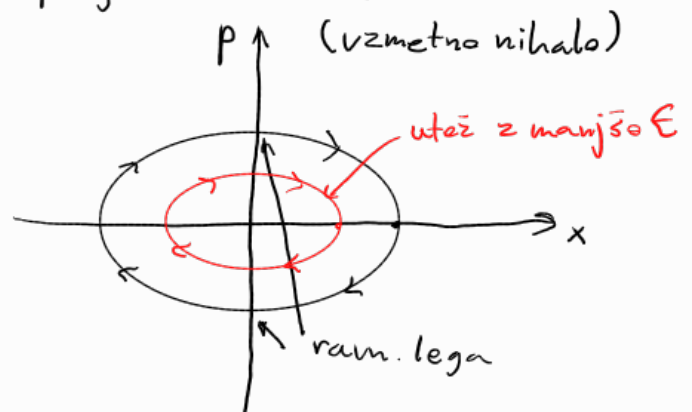
■ - delec 1

■ - delec 2

zgled: 1-D harmonični oscilator

x : koordinata

p : gibalna količina



$$x = x_0 \cos \omega t$$

$$p = -x_0 m \omega \sin \omega t$$

- čas. razvoj sistema = tir v faznem prostoru

$$\rho(p_1, p_2) = \rho(E(p_1), E(p_2))$$

$$\rho = \rho(E)$$

Povprečje

• opazljivka Y

$$\bar{Y} = \int Y \cdot \rho(E) d\Gamma \quad \text{diferencial faznega prostora}$$

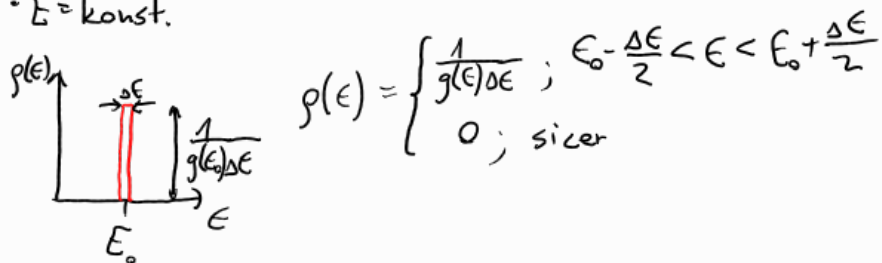
↑
verj. gost.

$$= \int Y \rho(E) g(E) dE$$

↑
gostota stanj

Mikrokanonična porazdelitev (ansambel)

• $E = \text{konst.}$



Kanonična porazdelitev

• delci/podsystemi so šibko sklopljeni

$$\boxed{\begin{matrix} 2 \\ 1 \end{matrix}} \begin{matrix} p_2(\epsilon_2) \\ p_1(\epsilon_1) \end{matrix} \quad p_{1+2}(\epsilon_{1+2}) = p_1(\epsilon_1) p_2(\epsilon_2)$$

$$\ln p_{1+2} = \ln p_1 + \ln p_2$$

$$\frac{\partial}{\partial \epsilon_1} \rightarrow \frac{1}{p_{1+2}} \frac{dp_{1+2}}{d\epsilon_{1+2}} \underbrace{\frac{\partial \epsilon_{1+2}}{\partial \epsilon_1}}_{=1} = \frac{1}{p_1} \frac{dp_1}{d\epsilon_1} = \text{funkc. } \epsilon_1$$

$$\frac{\partial}{\partial \epsilon_2} \rightarrow \frac{1}{p_{1+2}} \frac{dp_{1+2}}{d\epsilon_{1+2}} \underbrace{\frac{\partial \epsilon_{1+2}}{\partial \epsilon_2}}_{=1} = \frac{1}{p_2} \frac{dp_2}{d\epsilon_2} = \text{funkc. } \epsilon_2$$

• Ker sta desni strani funkc. razl. spr., bo enako le, če je to neka konst.

$$\Rightarrow \frac{1}{p} \frac{dp}{dE} = -\beta = \text{konst.}$$

$$\frac{dp}{p} = -\beta dE$$

$$\ln p = -\beta E$$

$$\rightarrow p(E) \propto \exp(-\beta E)$$

$$p(E) = C \exp(-\beta E)$$

↑
kasneje

Fazna vsota

= integral verj. gost. po vsem faznem prostoru

• rešitev statistično-fizikalnega problema

$$e^{-PF} = \int p(E) d\Gamma = C \int \exp(-\beta E) d\Gamma$$

$$1 = C \int \exp(-\beta(E-F)) d\Gamma$$

Povprečje

$$\begin{aligned} \bar{Y} &= \int Y p d\Gamma = C \int Y \exp(-\beta(E-F)) d\Gamma = \\ &= C e^{\beta F} \int Y e^{-\beta E} d\Gamma = \frac{C \int Y e^{-\beta E} d\Gamma}{e^{-\beta F}} = \\ &= \frac{\int Y e^{-\beta E} d\Gamma}{\int e^{-\beta E} d\Gamma} \end{aligned}$$

Povprečna E:

$$\bar{E} = \frac{\int E e^{-\beta E} d\Gamma}{\int e^{-\beta E} d\Gamma}$$

$$\frac{d(\beta F)}{d\beta} = \bar{E}$$

parameter integrala

$$\beta F = -\ln(C \int e^{-\beta E} d\Gamma)$$

Idealni plin

N delcev, 3D prostor

$$d\Gamma = dx_1 dy_1 dz_1 dx_2 \dots dp_{x1} dp_{y1} dp_{z1} dp_{x2} \dots$$

$$E = \frac{\vec{p}_1^2}{2m} + \frac{\vec{p}_2^2}{2m} + \dots$$

$$e^{-\beta F} = C \int \exp(-\beta (\frac{p_{x1}^2}{2m} + \frac{p_{y1}^2}{2m} + \dots)) d\Gamma =$$

$$= C \underbrace{\int dx_1 dy_1 dz_1 dx_2 \dots}_{V \dots} \underbrace{\int_{-\infty}^{\infty} \exp(-\beta \frac{p_{x1}^2}{2m}) dp_{x1}}_{\sqrt{\frac{2\pi m}{\beta}}} \underbrace{\int_{-\infty}^{\infty} \exp(-\beta \frac{p_{y1}^2}{2m}) dp_{y1}}_{\sqrt{\frac{2\pi m}{\beta}}} \dots =$$

$$= C V^N \left(\frac{2\pi m}{\beta} \right)^{3N/2}$$

$$\int_{-\infty}^{\infty} \exp(-ax^2) = \sqrt{\frac{\pi}{a}}$$

$$\rightarrow \beta F = -\ln C - N \ln V - \frac{3N}{2} \ln(2\pi m) + \frac{3N}{2} \ln(\beta)$$

$$\bar{E} = \frac{d(\beta F)}{d\beta} = \frac{3N}{2\beta} \xrightarrow{\beta = \frac{1}{k_B T}} \boxed{\bar{E} = \frac{3N}{2} k_B T = U}$$

$$\rightarrow C_V = \frac{d\bar{E}}{dT} = \frac{3}{2} N k_B, \quad C_V = \frac{C_U}{n^*} = \frac{3}{2} \frac{N}{n^*} k_B = \frac{3}{2} \frac{N_A}{M} k_B = \frac{3}{2} \frac{R}{M}$$

Ekviparticijski izrek: Vsaki neodv. neomejeni kvadratični prost. st. ustreza $\frac{1}{2} k_B T$

$$E = \dots + a_1 q_1^2 + a_2 q_2^2 + \dots$$

$$e^{-\beta F} = C \int \exp(-\beta E) d\Gamma = C \dots \int_{-\infty}^{\infty} \exp(-\beta a_1 q_1^2) dq_1 \int_{-\infty}^{\infty} \exp(-\beta a_2 q_2^2) dq_2 = C \dots \sqrt{\frac{\pi}{\beta a_1}} \sqrt{\frac{\pi}{\beta a_2}}$$

$$\bar{E} = \frac{d}{d\beta} \left(-\ln C - \ln \dots - \frac{1}{2} \ln \frac{\pi}{\beta a_1} - \frac{1}{2} \ln \frac{\pi}{\beta a_2} \right) = \dots + \frac{1}{2} \frac{1}{\beta} + \frac{1}{2} \frac{1}{\beta} = \dots + k_B T$$

Fluktuacije E

$$\sigma_E^2 = \overline{(E - \bar{E})^2} = \overline{E^2 - 2E\bar{E} + \bar{E}^2} = \overline{E^2} - 2\bar{E}\bar{E} + \bar{E}^2 = \overline{E^2} - \bar{E}^2$$

$$\boxed{\sigma_E^2 = -\frac{d^2 \beta F}{d\beta^2}} = -\frac{d}{d\beta} \bar{E} = -\frac{d}{dT} \bar{E} \cdot \underbrace{\frac{dT}{d\beta}}_{= 1/k_B T^2} = k_B T^2 \underbrace{\frac{d\bar{E}}{dT}}_{C_V \propto N} \Rightarrow \sigma_E \propto \sqrt{N}$$

$$\frac{\sigma_E}{\bar{E}} \propto \frac{1}{\sqrt{N}}$$